

Use 7Bchecks.iso to troubleshoot Bugcheck 7B_Windows_Tools_RDP SSH

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Tags

cw.TSG

cw.RDP-SSH

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Contents

- [When to use the tool:](#)
- [How to use the tool:](#)
- [How to interpret the log file:](#)
- [Additional Information:](#)
- [Escalation](#)
- [Need additional help or have feedback?](#)

When to use the tool:

The tool is mainly designed for help with troubleshooting/isolating No boot issues with bugcheck 7B (Thanks Adrian Zamfirescu for creating the tool). 0X7B means INACCESSIBLE BOOT Device. It is a Windows crash issue which happens when Windows tries to start, but fails to create/access the OS Disk/Volume or is unable to load any of the boot drivers for some reason (missing or corrupt driver, driver disabled in registry, etc.).

However, what makes 0x7B difficult to troubleshoot is:

1. Because of 0x7B, Windows fails to boot, we can only check the system in offline mode. Unlike online mode, in which we can check what the failed device is and what drivers it requires, in offline mode, what we have is only registry hive file (SYSTEM) and system files. We need to understand structure of Windows device registry and driver store to understand what device is failing and what configuration/driver the device is using.
2. Although it is a 0x7B bugcheck/crash, there is often no dump file (this is an interesting paradox: you need to write dump data to the disk upon crash, but currently, this kind of crash means it is inaccessible.).

How to use the tool:

Here is how it works:

1. Create a rescue VM environment by following steps from the article <https://learn.microsoft.com/en-us/troubleshoot/azure/virtual-machines/troubleshoot-vm-by-use-nested-virtualization> and install Hyper V role on the rescue VM.
2. Download the iso from here and share with the customer [7BChecks](#) . Please ask the customer to copy the iso to the Rescue VM.
3. Create a VM in Hyper V using the affected VM's OS disk (attached as a data disk on the Rescue VM). Mount the .iso file to the VM created in Hyper V, and boot from it
4. Once the WinPE boots up the tool will start automatically and will try and detect the OS installation on the system:

File Action Media Clipboard View Help



C:\> X:\windows\system32\cmd.exe - startnet.cmd

```
X:\windows\system32>wpeinit
```

```
X:\windows\system32>7bChecks
```

```
Detected the following Windows installations:
```

```
1 - D:\Windows
```

```
Select the one to use:
```

5. If it does not find the installation it will ask you to manually input it

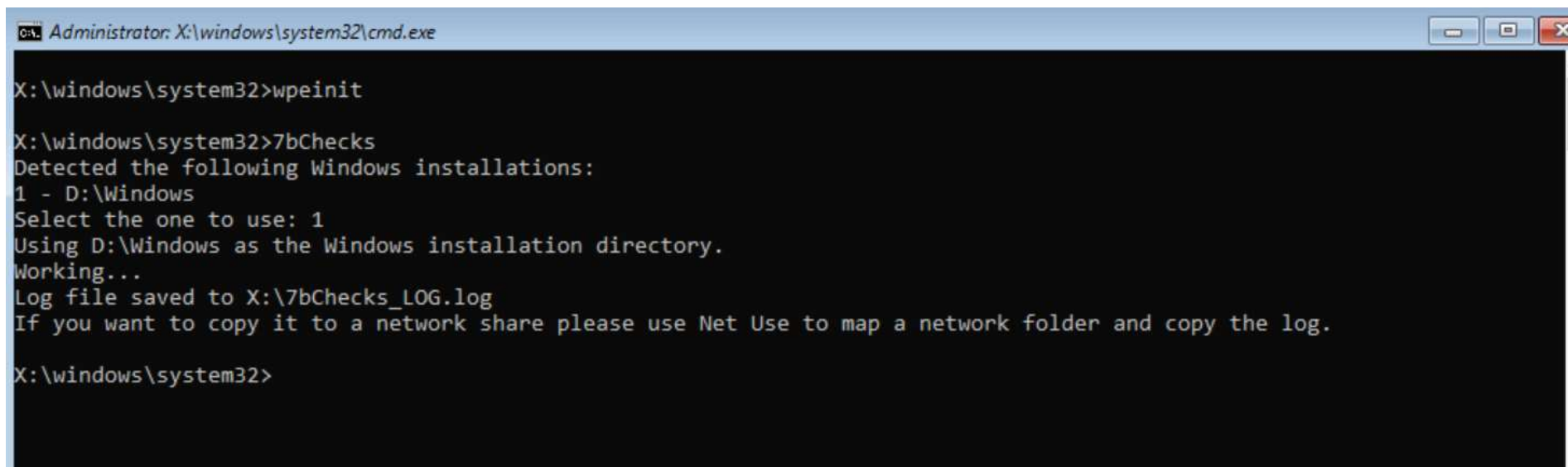
C:\> Administrator: X:\windows\system32\cmd.exe - 7bChecks.exe

```
Z:\>7bChecks.exe
```

```
No Windows installations detected.
```

```
Please type the path to the Windows directory (ex. C:\Windows):
```

6. Select the Windows installation to use or input it manually if the app cannot find it



```
Administrator: X:\windows\system32\cmd.exe

X:\windows\system32>wpeinit

X:\windows\system32>7bChecks
Detected the following Windows installations:
1 - D:\Windows
Select the one to use: 1
Using D:\Windows as the Windows installation directory.
Working...
Log file saved to X:\7bChecks_LOG.log
If you want to copy it to a network share please use Net Use to map a network folder and copy the log.

X:\windows\system32>
```

7. The app will run a script which will do a series of checks on the affected VM's OS disk:

1. Loads the SYSTEM hive of the OS. Checks all the services that are set to start at boot (drivers with Start type as 0).
2. Checks to see if the corresponding .sys files are present in Windows\System32\drivers.
3. Checks to see if DisableLKG key is present
4. Checks to see if DisableCDDDB and DontStartRawDevices are present
5. Checks for pending.xml (if any updates are in pending state)
6. Unloads the registry hive
7. Runs dism /image:c:\ /Get-Packages and shows the list of updates
8. Runs bcdedit /enum /all to show BCD entries
9. Runs diskpart list disk
10. Runs diskpart list volume

8. Once it finishes it will save a log in X:\7bChecks_LOG.log

The log will look similar to [7bchecks log file](#)

How to interpret the log file:

SERVICES CHECKS section lists the drivers which are set to start during boot (drivers with Start type 0) and checks if the driver is present or not. Please validate that the boot drivers are present.

If you observe an error similar to: Service tempOfflineHive\ControlSet001\Services\CLFS has Start type 0. Driver D:\Windows\System32\drivers\CLFS.sys NOT found

Please check if a copy of the driver is present in C:\Windows\System32\DriverStore\FileRepository folder. If yes, please copy the driver. We can also run SFC scan to fix file corruption, or copy the driver from a system with same OS and similar patch level.

[Please ignore if you see below errors in the logfile:

Service tempOfflineHive\ControlSet001\Services\Fs_Rec does NOT have a value ImagePath

Service tempOfflineHive\ControlSet001\Services\Wof does NOT have a value ImagePath]

FILTER DRIVERS CHECKS check if any third party filter drivers are present. If they are, please try removing the third party filter drivers from the Upperfilter/LowerFilter entry and check if the VM boots up.

DISM checks lists the updates that are installed on the VM and lists if there are any updates in Pending or Staged state. We can try reverting/removing the updates to try booting the VM.

Additional Information:

[How to use 7Bchecks.iso](#)

[Advanced Troubleshooting for inaccessible Boot device](#)

Escalation

1. If this doesn't work out, please reach out to the [Unable to RDP-SSH SME channel on teams](#) for advise providing the case number, issue description and your question

Need additional help or have feedback?

<i>To engage the Azure RDP-SSH SMEs...</i>	<i>To provide feedback on this page...</i>	<i>To provide kudos on this page...</i>
If you are in the OneVM team then reach out to the RDP-SSH SMEs AVA channel via Teams. If you are a VM Delivery Partner then follow your team's process for swarming or ask your TA. Make sure to use the Ava process for faster assistance.	Use the RDP-SSH Feedback form to submit detailed feedback on improvements or new content ideas for RDP-SSH. Please note the link to the page is required when submitting feedback on existing pages! If it is a new content idea, please put N/A in the Wiki Page Link.	Use the RDP-SSH Kudos form to submit kudos on the page. Kudos will help us improve our wiki content overall! Please note the link to the page is required when submitting kudos!